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TCN-BiLSTM-CE: An interdisciplinary approach for missing energy data imputation by contextual inference

Mengbo Yu*, Alexander Neubauer, Stefan Brandt, Martin Kriegel

Hermann-Rietschel-Institut, Technische Universität Berlin, 10587 Berlin, Germany

HIGHLIGHTS

- Cross-domain methods are used for image filling to adapt models suitable for filling missing values in energy demand data.
- Introduction of Temporal Convolutional Network and Bi-directional LSTM enhances applicability to contextual information extraction of energy data.
- The reconstruction loss and the adversarial loss are used during the network training for calculating the loss with and without missing values, respectively.
- Missing data of different energy types, different types of missing values (continuous, discrete) in different periods were reconstructed and evaluated separately.
- Model for missing energy data imputation adapted from image-filling model outperformed LSTM models, SOTA models and original image model.

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ABSTRACT

Energy data frequently contain missing values, which complicate accurate analysis and informed decision-making. Since energy load data are generally in the form of time series, the problem of reconstructing missing data in energy load data can also be translated into the problem of reconstructing missing data in time series data. Inspired by image-based techniques, this study introduces a novel approach for imputing missing energy data using a Context Encoder (CE), in order to translate the original network from the task of reconstructing the missing parts in image data to the task of reconstructing missing data in time series data, using the complete data before and after the time period in which the missing data appears as the contextual reference to infer the missing data in the missing time period. Two structures commonly used in time series data processing, TCN and Bi-LSTM, were introduced to learn the latent representation in latent space, in order to extract the features of bidirectional contextual information, causal relationships and long-term dependency in time series data. After conducting separate experiments for electrical load data and heat load data, we demonstrate the effectiveness of our proposed method on energy data for different energy types. Compared to other baseline methods, the Normalized Root Mean Squared Error (NRMSE) between original and reconstructed electric load data and heat load data reduced from 7.58 %–81.20 % and 4.26%–65.74%, respectively, fully demonstrating the potential of using the proposed method for reconstructing the missing energy load data.

1. Introduction

Energy data encapsulate vital information essential for analyzing and managing the complex dynamics of energy systems. From the production and consumption to distribution and efficiency of energy resources across diverse sectors and domains, energy data serve as a vital cornerstone, playing a pivotal role in shaping policies, driving innovation, and fostering sustainable development. Energy data reveal usage and load patterns across sectors such as residential, commercial, industrial, and transportation, offering critical insights. These insights aid decision-makers in analyzing consumption trends, identifying inefficiencies, and forecasting future loads. Armed with this knowledge, strategies can be devised to promote conservation, enhance reliability, and reduce costs, thus optimizing resource utilization and bolstering sustainability efforts.

Effective management and analysis of energy data are critical to ensure the efficiency and sustainability of modern energy systems, however, real-world energy datasets often suffer from missing values due to various factors such as sensor failures, communication errors, or data

* Corresponding author at: TU Berlin, Marchstr. 4, 10587 Berlin, Germany.
Email address: mengbo.yu@tu-berlin.de (M. Yu).

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Nomenclature

Abbreviations

ARIMA	AutoRegressive Integrated Moving Average
Bi-LSTM	Bidirectional LSTM
BRITS	Bidirectional Recurrent Imputation for Time Series
CE	Context Encoder
CNN	Convolutional neural network
FC	Fully Connected Layer
GAN	Generative Adversarial Network
IQR	Inter Quartile Range
k-NN	k-Nearest Neighbour
LI	Linear Interpolation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAR	Missing at Random
MBE	Mean Biased Error
MCAR	Missing Completely at Random
MLP	Multi Linear Regression
MRNN	Multi-Directional Recurrent Neural Networks
MSE	Mean Squared Error
NMAR	Not Missing at Random
NRMSE	Normalised RMSE
RMSE	Root Mean Square Error
SOTA	The state of the art
SVM	Support Vector Machine
TCN	Temporal Convolutional Network

Greek symbols

λ	Weight hyperparameter
σ	Activation function
θ	Model weight

Latin symbols

b	Bias term
d	Dilation factor
h_t	Output of Bi-LSTM at time t
K	Kernel size
$w(k)$	Convolutional filter weight at lag k
W_b	Weight of the backward hidden state
W_f	Weight of the forward hidden state
$x(t)$	Input at time t
$\mathbf{M}$	Binary mask
$\mathbf{x}$	Original data
i	i-th training epoch
L	Model loss
n	Numbers of data point
T	Training epoch
x	Data point

collection constraints [1]. Given an energy data with missing values, it is critical for us to accurately reconstruct these missing values for further accurate analysis, forecasting and decision making in energy management. The energy load data we used in this paper came from smart meter measurements, also presented as time series data [2]. Therefore, we transformed this problem into the problem of reconstructing missing data in time series data and conducted a literature review of studies on missing data reconstruction in energy load data that were also in the form of time series data, analyzed their limitations, and proposed our network to address the research gap.

1.1. Literature review

Methods for reconstructing missing energy load data have been extensively investigated in previous studies. Commonly used methods can be divided into two categories, statistical methods and machine learning methods. Historically, statistical methods have been widely employed for missing energy data reconstruction task. These methods include Auto Regressive Integrated Moving Average (ARIMA) [3], Expectation-Maximum [4], Multivariate Imputation by Chained Equations (MICE) [5] and Linear Interpolation (LI) [6]. Although statistical methods can effectively reconstruct the missing energy data, machine learning methods have proven to be a better choice than statistical methods in most scenarios. In their comprehensive study on filling in missing building sensor data, Chong et al. [7] demonstrated the superior performance of machine learning methods such as Linear Regression (LR), k-Nearest Neighbour (k-NN), and Support Vector Machine (SVM) over statistical methods like mean imputation and zero substitution. Another notable comprehensive study comparing statistical and machine learning methods was conducted by Kim et al. [8]. In their study, the reconstruction effect for residential building energy data between statistical methods such as mean imputation, LI and machine learning methods such as random forest, SVM and Multi Linear Regression (MLP) were evaluated, which underscored the superiority of machine learning methods in imputing missing data. In the case study on imputing missing electric power data from an electric power station in Taoyuan City in Taiwan, Wang et al. [9] divided the reconstructed electric power data into summer and non-summer data and also observed that in terms of electric current, machine learning methods such as k-NN, SVM and MLP showed a lower mean absolute percentage error 17.17%–21.46% compared to statistical methods such as ARIMA and LI 23.5%–31.65% in summer. Similarly, machine learning methods showed errors of 16%–18.39% compared to statistical methods 17.87%–28.72% in non-summer. In recent years, there has also been significant research into machine learning techniques specifically for time series data, such as using Long Short-Term Memory (LSTM) and Bidirectional LSTM (Bi-LSTM), which has led to improvements. Ma et al. [10] demonstrated the efficacy of LSTM networks in filling missing energy data, showcasing superior performance compared to other methods under different missing rates in energy data such as k-NN and SVM. Meanwhile, Jiang et al. [11] implemented a Bi-LSTM network for the reconstruction of missing energy data across varying durations of missing hours, demonstrated that the Bi-LSTM approach outperformed other methods, exhibiting lower Root Mean Square Error (RMSE) and Mean Absolute Error.

With the development of machine learning techniques, some of the state of the art (SOTA) methods dedicated to time series missing data reconstruction have been widely used, such as Bidirectional Recurrent Imputation for Time Series (BRITS) [12] and Multi-Directional Recurrent Neural Networks (MRNN) [13]. Moreover, recent studies have also emerged that draw on cross-domain methods to implement in cases where energy load data also appears in the form of time series data. Fu et al. [14] have shown the effectiveness of reconstructing the missing energy load data through a cross-domain approach using U-Net [15], which is commonly used in the field of computer vision. Converting the time series data format to image data format to fill in the missing values significantly outperforms the other methods mentioned in the original paper: 1D-CNN, 2D-CNN, and the statistical method weekly persistence.

1.2. Limitations of previous studies and our purpose

However, both BRITS and MRNN, although using contextual information to infer missing data, have to fill missing data in time series data by point-by-point estimation. This inevitably leads to the use of data that has already been estimated by the model during the filling process, which will deviate more and more from the ground truth due to the accumulation of errors when reconstructing long term missing data. Moreover, since the missing values are filled point-by-point, the long-term dependencies of the time series are not well incorporated into the missing data reconstruction process. In addition, it has not been widely studied whether these SOTAs can also perform well in reconstructing the missing data in energy load data. Regarding missing data reconstruction using methods from other disciplines, as in Fu et al. [14], he only converted all the building energy data in the dataset into image data to train the model, and did not take into account the differences between different buildings and energy load patterns. In addition, directly using the original model for the image data task would also fail to learn the specific characteristics of the energy load data presented as time series data.

In order to overcome the above mentioned limitations and achieve better effectiveness, in this paper, we propose our network TCN-BiLSTM context encoder, for reconstructing the missing data in energy load data by contextual inference. Like the study by Fu et al. [14], our approach was also inspired by its use in the task of missing image data reconstruction, which inferred the missing parts from the information of the pixels around them [16–19]. Similar to the original approach, our network also uses the past and future observations around the missing energy load data as contextual prompts to infer missing energy load data in the corresponding time period. To the best of the author's knowledge, there are not many studies on solving the missing data reconstruction task in time series data through contextual inference. This method of contextual inference is meaningful to be utilized as a solution for the missing data reconstruction in energy load data that appears as time series data, because energy load data in the form of time series tends to have regular periodicity patterns [20,21]. As long as the general causal relationship and long-term dependency are learned, the missing data can be accurately inferred based on the data before and after the missing part as a contextual prompt. The idea of analogy to the application of missing data filling in image data is to divide all the energy load data in a time period of single building into a set of data with the fixed time length, regard them as a set of 'image data' with similar patterns and pixel values, learn the similar relationship between the preceding and following texts, and fill in the long-term missing data that occur in the measurement process later on.

To capture the contextual features of the time series, the causal relationships between data corresponding to adjacent time steps, and the long-term dependencies of the entire time series, we introduced TCN and Bi-LSTM, a pair of modules often used in solving the problems of time series data prediction [22–24], into our unsupervised learning-based model to extract the corresponding features in the energy load data appearing as time series data. Bi-LSTM was used to extract features related to context [25], while TCN was used to extract the causal relationships and the long-term dependencies in the energy load data. A set of hybrid latent representations of the time series data features is then formed, which enables the decoder to better infer the missing data based on this set of latent representations using the prompts of the complete data in the previous and future data. In order to make the reconstructed data match the pattern of the original data, other features that could affect the pattern of original energy load data, such as temperature, time of day, day of week, were also included in our input data, so that the reconstructed data are not only numerically close, but also match the data pattern of original energy load data.

The novelty of this work lies in a unique cross-domain approach that extends the method from missing image data reconstruction to missing energy data, highlighting the potential of cross-domain methods in solving energy data problems. By using a hybrid TCN-BiLSTM model within an unsupervised learning framework, the temporal dependencies with contextual information of long-term missing data in time-series energy data are better integrated than directly using the original model, which improves the feature extraction capabilities for energy data appearing as time-series data and achieves the goal of accurately inferring long-term missing data by using the complete data before and after the missing part as a contextual prompt. As a more concise summary of the intent of this paper, the main contributions of this paper are described as follows:

1. The effectiveness of a not widely used method for reconstructing missing energy load data using contextual inference is explored.
2. Potential of TCN and Bi-LSTM, a combination often used for prediction of time series data, when applied to unsupervised learning for time series feature extraction is explored.
3. Potential and effectiveness of the state of the art methods for missing data reconstruction of energy data were also explored during the comparison with our proposed TCN-BiLSTM-CE.
4. Cross-domain approach was attempted by implementing the model for imputing missing part of image data in missing energy data reconstruction.

The remainder of this paper is structured as outlined below. Section 2 introduces the TCN-BiLSTM-CE which is dedicated to the missing energy data reconstruction task. Section 3 presents the dataset used for the experiment, the data pre-processing method, and the approach for generating the samples from the original energy data. The experimental results are provided in Section 4, and the conclusions are drawn in Section 5.

2. Methodology

2.1. Overview of proposed network

The structure of our approach is based on the CE developed by Iizuka et al. [19] for solving the image data imputation task, which allows obtaining locally and globally consistent images when filling in missing regions of arbitrary shape to complement images of arbitrary resolution. Fig. 1 shows the architecture of the whole framework. The architecture begins with an input layer that receives incomplete energy data, which contain missing values. Similar to the original CE network, TCN-BiLSTM-CE architecture also consists of three networks: a completion network, a global context discriminator, and a local context discriminator. The completion network serves as generator to complete the missing energy data, while both the global and the local context discriminators are auxiliary networks used exclusively for training. Both generator and discriminator will be further introduced in Sections 2.1.1 and 2.1.2 respectively. In order to adapt the network to process energy data in the form of time series, the convolutional layer is changed from 2D to 1D. As we described in Section 1.2, input data is the data combined with the original energy load and other variables, such as weather data and temporal features, in order to simulate the correct pattern of energy load data.

2.1.1. Generator

Generator, also known as completion network, is used to enable reconstruction of missing energy load data through contextual inference, consisting of multilayer CNNs, TCNs and Bi-LSTMs. Encoder and decoder in generator consist of CNNs, while the feature extraction part consists of TCNs and Bi-LSTMs. Like the original network in [19], the encoder of the TCN-BiLSTM-CE also adopts the same structure as the first six layers of AlexNet [26], which extracts temporal features of energy load data and reduces data dimensions to avoid the impact of the "curse of dimensionality" [27] on the extraction of relevant temporal features by Bi-LSTM and TCN. Once the dimensionality of the original data has been reduced, the feature extraction layer consisting of TCN and Bi-LSTM can proceed with features related to contextual inference, i.e., extracting causality and long-term dependencies through the TCN layer, and extracting feature representations of contextual relationships in time series data through the Bi-LSTM layer. Combining the TCN and Bi-LSTM layers as one feature extraction module allows causal relationships, long-term dependencies, and contextual relationships to be combined into a set of hybrid latent representations that better integrate these features in the context of contextual inference. In the feature extraction part, we linked two sets of this module to better extract temporal features. Based on this

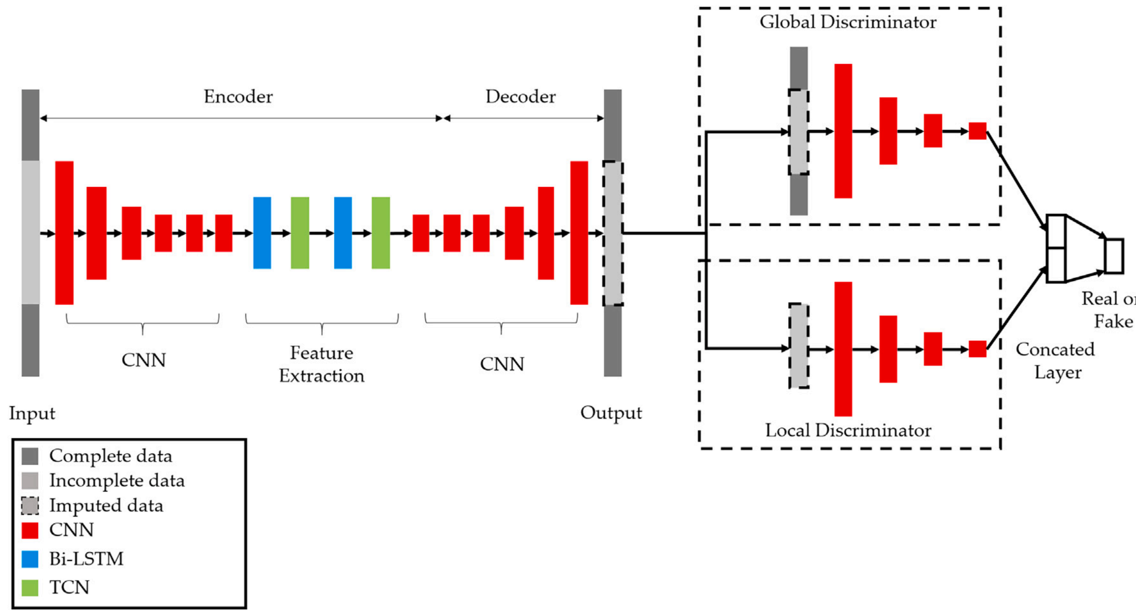

Fig. 1. TCN-BiLSTM-CE framework.

set of latent representations, a decoder with the same structure as the encoder is finally used to reconstruct the missing data as closely as possible to the ground truth using the contextual information of the complete data before and after the time period corresponding to the missing data, utilizing the contextually relevant feature representations learned from the feature extraction layer.

2.1.2. Discriminator

To verify that the reconstructed results from the generator match the values and patterns of the original energy data, the discriminator is included in the network training process as another part of the overall TCN-BiLSTM-CE. Following the structure of the original model, the TCN-BiLSTM-CE consists of a global discriminator and a local discriminator, which focus on the different aspects: the global discriminator focuses on discriminating the whole energy load data after adding the reconstructed data to the original data, in order to determine whether the filled data matches the global context, thus minimising cases such as large offsets despite similar shapes. The local discriminator, on the other hand, only discriminates the missing parts generated by the generator to determine whether the reconstructed data matches, for example, the periodicity pattern and the upward or downward trend of the original data.

As in the case of the adversarial training network described in [28], in our network, the generator and the discriminator collaborate jointly in the same way: During each training iteration, the energy load data after being filled through the generator will be fed into the discriminator, which will discriminate the generated data by deriving the probability of whether the generated data came from the ground truth. Once the discriminator is trained after each training iteration, the generator will be updated accordingly, in order to maximize the probability from the discriminator and therefore, make the generated results from the generator as close as possible to the ground truth.

2.2. Introduction of the networks in the proposed framework

2.2.1. Convolutional neural network

As we have described in Section 2.1, three main networks are included in our proposed framework: CNN, TCN and Bi-LSTM. The first part of the network that will be introduced is CNN. CNN is a popular discriminative deep learning architecture that learns directly from the input without the need for human feature extraction [29]. CNNs have achieved remarkable success in image processing tasks due to their ability to exploit the spatial information of images, which makes the application of CNNs to time series data equally relevant, since the correspondence between different time points and energy loads, and the shape changes of the time series due to periodicity have some similarity to the spatial information of images [30]. Based on this feature, CNN can effectively capture the temporal information features based on time series data while reducing the dimensionality of the data.

2.2.2. Temporal convolutional network (TCN)

Another network that will be introduced in TCN-BiLSTM-CE framework is TCN. TCN is a type of convolutional neural network with a special design that makes it suitable for handling time series data. It includes two parts that can fulfill our purposes, i.e., causal convolution and dilated convolution. Causal convolution can extract the causal relationship between each data point in time series data, while dilated convolution allows interval sampling of the convolution input, thereby achieving the goal of being able to capture longer-term patterns [31]. Both features are critical for making inferences based on the given past and future information, so we use this module as part of the feature extraction [32]. Besides, the residual connection in TCN block can also improve performance in very deep architectures [33], and consists of adding the input of a TCN block to its output. Eq. (1) shows the mathematical expression of the collaborative structure of TCN, where $o(t)$ is the output at time t, $x(t)$ denotes the input at time t, $w(k)$ the convolutional filter weight at lag k, K the kernel size, d the dilation factor, b the bias term, and σ the activation function.

$$o(t) = \sigma\left(x(t) + \sum_{k=0}^{K-1} w(k) \cdot x(t - d \cdot k) + b\right) \quad (1)$$

2.2.3. Bi-directional long short memory (Bi-LSTM)

Due to the issues such as exploding or vanishing gradients when dealing with long sequences with RNN, LSTM was invented by Ref. [34] to solve this issue. By introducing the mechanisms of input gate, forget gate, and output gate, the problems of gradient vanishing and exploding present in RNNs are addressed, and they also allow them to better capture and maintain long-term dependencies within sequential data. As an advanced form of LSTM, Bi-LSTM incorporates a bidirectional topology that allows it to process data over the full time domain, which enables both the use of previous features and the inclusion of future information. This mechanism of Bi-LSTM can effectively extract the feature representation of contextual information, and this just compensates for the defect that TCN networks cannot effectively extract the bidirectional semantics of time series [35]. Eqs. (2)–(4) show the mathematical expression of the collaborative structure of Bi-LSTM, where h_t is the final Bi-LSTM output at time t, computed by applying the activation function σ to the weighted sum of the forward hidden state h_{t+1} with weight W_f, the backward hidden state h_{t-1} with weight W_b, plus the bias term b.

$$\overrightarrow{h_t} = LSTM_{forward}(x_t, \overrightarrow{h_{t-1}}) \tag{2}$$

$$\overleftarrow{h_t} = LSTM_{backward}(x_t, \overleftarrow{h_{t+1}}) \tag{3}$$

$$h_t = \sigma\left(W_f \cdot \overrightarrow{h_t} + W_b \cdot \overleftarrow{h_t} + b\right) \tag{4}$$

2.3. Network training

In this section, the training procedure of TCN-BiLSTM-CE network will be introduced. Inspired by the training procedure in [19], we modified the training process accordingly for our application scenario, making our model more extensible to accommodate different data patterns for different types of energy owned by different buildings. During the whole training procedure of the network, generator network $G(\cdot)$ is trained first alone, after that, the discriminator $D(\cdot)$ is trained together with $G(\cdot)$, so as to achieve the purpose of minimising the loss of the network. The training process is described as follows. Firstly, $G(\cdot)$ is trained for t_G epochs. After t_G epochs of training, $D(\cdot)$ is added to the training procedure and trained with the pre-trained generator $G(\cdot)$ together for t_D epochs, in order to minimize the losses of the whole network. The training procedure was designed in this way, in order to allow the pre-trained generator to produce some meaningful outputs to serve as input to the discriminator, ensuring a stable starting point for the adversarial training process and avoiding the instability and potential mode collapse caused by training with random initialization, which are the common problems encountered with other generative network methods that generate samples from noise. The discriminator, initialized from scratch, started as a randomly initialized network, which allowed the discriminator to dynamically adapt to the outputs of the generator and provided constructive gradients to guide the improvements of the generator after each iteration of the adversarial training process. By starting from scratch, the discriminator avoids any pre-trained bias and focuses entirely on counteracting the outputs of the generator throughout the training process. Together, this setup fosters a competitive adversarial process in which the pre-trained generator and the scratch discriminator iteratively improve each other's performance. In order to validate, whether it is the optimized training procedure in our case, two other training procedures were also applied as baselines, i.e., training without discriminator and training the generator and discriminator at the same time.

As in original model that was proposed by Iizuka et al. [19], TCN-BiLSTM-CE consisted of two parts of losses that needed to be calculated, a weighted Mean Squared Error (MSE) loss for training stability, and a Generative Adversarial Network (GAN) loss to improve the realism of the results. Therefore, a joint reconstruction loss to calculate both the reconstruction loss and the adversarial loss is envisaged in this framework. Let G($\mathbf{x}$,M_G) and D($\mathbf{x}$,M_D) represent generator and discriminator of TCN-BiLSTM-CE framework as functional forms, respectively, where $\mathbf{x}$ is the original data without any missing values, M_G and M_D are two binary matrices used to indicate the presence or absence of missing values for each input vector. The binary masks M_G and M_D take the value 1 when the value is not missing and 0 when the value is missing. MSE loss that is used to stabilize the training is defined in Eq. (5). We randomly created continuous missing values from 4 to 10 consecutive days in the original data by setting the value in M_G and M_D to null. $G(\mathbf{x} \odot M_G)$ represents the data that were processed by generator $G(\cdot)$. By comparing the original data and the data processed by the generator, the reconstruction loss L_{rec} is calculated.

$$L_{rec} = \|\mathbf{x} - G(\mathbf{x} \odot M_G)\|^2 \tag{5}$$

Another part of the loss in this framework is adversarial loss, which was first introduced in [28]. A discriminator D(·) is trained, in order to maximize the probability of assigning the correct label to both training examples and samples from G(·). G(·) is trained simultaneously to minimize the $\log(1 - D(G(\cdot))$. Therefore, the adversarial loss can be defined as:

$$\min_G \max_D \mathbb{E}[\mathbf{log} D(\mathbf{x}, M_D) + \mathbf{log}(1 - D(G(\mathbf{x} \odot M_G), M_G))] \tag{6}$$

By combining it with reconstruction loss, the loss function for the whole TCN-BiLSTM-CE framework can be written as:

$$\min_G \max_D \mathbb{E}[L_{rec}(\mathbf{x}, M_G) + \lambda[log D(\mathbf{x}, M_D) + log(1 - D(G(\mathbf{x} \odot M_G), M_G))] \tag{7}$$

where $L_{rec}(\mathbf{x}, \mathbf{M})$ is the reconstruction loss that is represented in Eq. (5), λ is the weighted hyperparameter. The joint loss of the TCN-BiLSTM-CE network is calculated and updated by Eq. (8) according to Eq. (7).

$$\mathbb{E}[\nabla\theta_G L_{rec}(\mathbf{x}, M_G) + \nabla\theta_D \lambda[log D(\mathbf{x}, M_D) + log(1 - D(G(\mathbf{x} \odot M_G), M_G))] \tag{8}$$

The entire training procedure to minimize the loss of the framework is shown in Algorithm 1, where T denotes the total number of batches trained on the network and i denotes the i-th training epoch conducted against the network.

2.4. Missing data type

According to Demirhan and Renwick [36], since the mechanism behind the missing data cannot be known exactly, the following three types of missing values are often summarized in energy load data in the form of time series: When there is no specific mechanism creating missing data, it is called Missing Completely at Random (MCAR). When there are other variables affecting the existence of missing values and likelihood of having a missing value is independent of the value itself, it is called Missing at Random (MAR). On the contrary, if the likelihood of having a missing value is associated with the missing value itself, it is called not Missing at Random (NMAR). In our approach, we do not need to know the causes

Algorithm 1 Training procedure TCN-BiLSTM-CE.

while $i < T_G + T_D$ **do**
  if $i < T_G$ **then**
    Sample a minibatch of training samples from training data $\mathbf{x}$.
    Generate masks M_G with random holes for each sample in a minibatch for the training procedure of the generator network G.
    Update the generator network G with the weighted MSE according to Eq. (5).
  else
    Generate masks M_D with random holes for each training data $\mathbf{x}$ in a minibatch for the training procedure of the discriminator network D.
    Update the discriminator network D with the adversarial loss according to Eq. (6).
    Update the TCN-BiLSTM-CE network with the joint loss according to Eq. (8).
  end if
end while

of missing values, whether they occur discretely or continuously over a long period of time, but simply fill in the missing data using contextual information between the energy load data and other variables related to changes in energy load as auxiliary contextual prompts. Therefore, our study can be described as missing data reconstruction for MAR.

2.5. Evaluation metrics

In order to more effectively measure the effectiveness of reconstruction of missing data for different types of energy data, three different metrics for evaluating the model performance, NRMSE, Mean Absolute Error (MAE) and Mean Biased Error (MBE) are used. The three metrics each provide a distinct perspective on imputation performance, capturing accuracy, variability, and bias respectively. MAE measures the average absolute error in the original data, providing a clear indication of typical deviation [37]. NRMSE normalises RMSE to make it comparable across different scales [38]. MBE captures systematic bias, indicating whether the model consistently over- or underestimates true values [39]. Eqs. (9)–(11) show the calculation of these three metrics respectively, where x_{max} and x_{min} represent the maximal and minimal values in time series X, n is the number of all points in time series, x_i and $\hat{x}_i$ are the real value and predicted value respectively.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|x_i - \hat{x}_i\right| \tag{9}$$

$$\text{NRMSE} = \frac{1}{x_{\text{max}} - x_{\text{min}}} \cdot \sqrt{\frac{\sum_{i=1}^{n}(x_i - \hat{x}_i)^2}{n}} \cdot 100\ \% \tag{10}$$

$$\text{MBE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i) \tag{11}$$

3. Data preparation and model setting

3.1. Introduction of experimental dataset

3.1.1. Electricity load dataset

Electricity load data comes from the external dataset, which is called *The Building Data Genome Project 2* [40], is used in our experiment. The dataset covers data on load for different types of energy (e.g., electricity, gas, hot water) for different types of buildings (e.g., retail, education, offices) from 01/01/2016 to 12/31/2017. Electricity load data from 12 buildings were selected as training and test data, where training data and validation data were selected from the time period 06/05/2016 – 12/31/2016 in this dataset with a percentage of 80% and 20% respectively, while the validation data for the electricity data, were selected from the time period 07/03/2017 to 07/26/2017.

As described in Section 1.2, in order to make the reconstructed missing data correspond to the load pattern under the influence of different features, it is also necessary to find the features that significantly influence the load change pattern and include them as part of the input data to the model. Table 1 summarizes the selected input features when reconstructing missing data in the electricity load data. Since the two features, day of the week and hour of the day, have already been used in previous research on electricity load data [41,42], this dataset, also shows a significant cyclical trend with the day of the week and the hour. This is illustrated as an example in Fig. 2 with the building named ‘Mouse health Estela’. The electricity load on weekdays and during the daytime is greater than that on weekends and during the nighttime hours, which further illustrates the importance of using these two variables as well as the electricity load data to accurately reconstruct the data pattern.

Table 1
Input features for reconstructing the missing data in electricity load data.

Features	Label
Electricity load data, hour, day of the week	$\dot{Q}_{\text{elec}}$

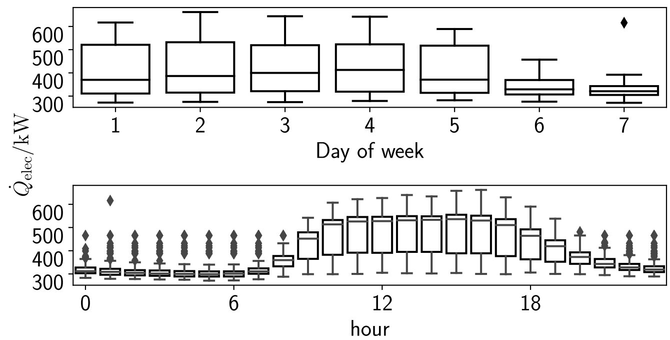

Fig. 2. Relationship between hours and days of the week with electricity load data.

Table 2
Input features for reconstructing the missing data in heat load data.

Features	Label
Heat load data, hour, temperatur	$\dot{Q}_{\text{heat}}$

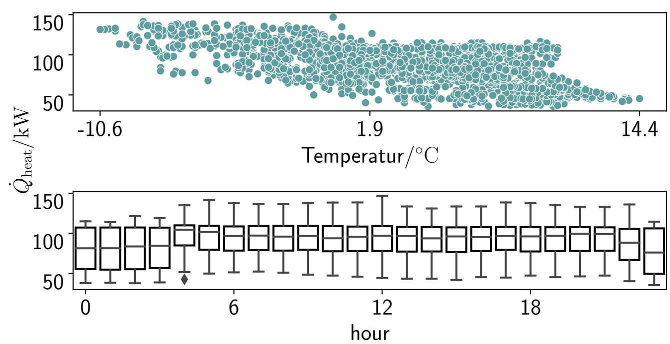

Fig. 3. Relationship between temperature and hours with heat load data.

3.1.2. Heat load dataset

The heat load data were selected from smart meter data measured by Green Fusion GmbH from 10/10/2023 to 04/07/2024 for a number of residential buildings in Berlin. Heat load data from 8 buildings that were measured from 11/01/2023 to 02/14/2024 were selected for the experiment as training and validation data with a percentage of 80% and 20% respectively. The test data were selected from 02/15/2024 to 03/04/2024. As in the study of reconstructing missing data for electricity load data, since the two features of temperature and time of day were also commonly used in the previous study for heat load data [43,44], these two features were also included in the input to reconstruct the pattern of heat load data. Table 2 summarizes the selected input features when reconstructing missing data in the heat load data. The relationship between heat load data and temperature as well as day of the hours is demonstrated in Fig. 3.

3.2. Sample modeling

After the preparation of training dataset, it is necessary to transform the collected data from a whole time series into the form that is suited to train a model to reconstruct the missing data. As a common way to handle the time series problem, rolling window is used in our experiment to

Original observation	x_1	x_2	…	x_{24}	x_{25}	…	x_{336}	x_{337}	…	x_{361}	x_{362}	…	x_n
Sample 1	x_1	x_2	…	x_{24}	x_{25}	…	x_{336}						
Sample 2					x_{25}	…	x_{336}	x_{337}	…	x_{361}			

Fig. 4. Creating samples with rolling window.

transform the whole series of historical data into different samples [45]. Assuming $X = [x_1, x_2, \dots, x_N]$ represents a time series consisting of N observations, which have the same time interval between each observation. To generate training samples via a rolling window approach, we define w as the window length and Δt as step size for each generated sample. By determining these two parameters, the whole time series will be finally partitioned into $N - (\Delta t + w) + 1$ samples. To clarify it more clearly, Fig. 4 shows the process of using rolling window to generate the first two samples with a visual form in our case. In our case, w and Δt are set to 336 and 24 h, which represent that each time sample contains up to 14 days of data, and once a sample is created, the sliding window moves back 24 h along the time step. The data is collected every hour and represents the hourly values of the electricity load data and heat load data. Using this method, the test data from the heat and electricity load data was split into 10 samples, which are reconstructed with missing data and the results are averaged as a basis for comparing the results.

3.3. Data preprocessing

In order to ensure the quality of the model after it has been trained, the data first needs to be preprocessed to ensure that it is of the highest possible quality before we put it into training. The data preprocessing method presented in this paper consists of filtering out outliers and data normalization.

3.3.1. Outlier filtering

In this paper, we determine whether the value is an outlier by calculating the Inter Quartile Range (IQR) of the energy load data for corresponding hour. Any value that lies outside the range of (25th Percentile − 1.5·IQR) to (75th Percentile + 1.5· IQR) is noted as outlier, where the values less than (25th percentile − 1.5· IQR) and greater than (75th percentile + 1.5· IQR) are substituted with (25th percentile − 1.5·IQR) and (75th percentile + 1.5·IQR) in this paper respectively [46].

3.3.2. Data normalization

To maintain the large variation of predictions the normalization technique is required to make the input data closer. Normalization is scaling technique a mapping technique or a pre-processing stage, where we can find new range from an existing range. In this paper we use Min-Max normalization as our method to normalize the original data [47]. Eq. (12) shows the principle of preprocessing with min-max normalization.

$$x_{i,norm} = \frac{x_i - x_{i,\min}}{x_{i,\max} - x_{i,\min}} \tag{12}$$

where $x_{i,\min}$ and $x_{i,\max}$ represent the minimal and maximal values in the original data, $x_{i,norm}$ represents the data after normalizing with min-max normalization.

3.4. Model setting

3.4.1. Experimental setting baseline models

In order to verify the performance of the TCN-BiLSTM-CE network, we need to compare the effectiveness of TCN-BiLSTM-CE for missing energy data reconstruction with the results obtained by other commonly used methods for missing energy data reconstruction. We selected both LSTM methods (Bi-LSTM, LSTM), SOTAs for missing data reconstruction (BRITS, MRNN, Transformer) as well as the original model proposed by Iizuka et al. [19] that was transformed from original 2D model to 1D model. A brief description of these methods is as follows:

- LSTM, Bi-LSTM: As discussed in Section 2.2.3, LSTM network is an enhanced RNN, which has been experimentally proven to have higher validation and prediction accuracy than the standard RNN for long-range dependent input and output data by effectively solving the gradient vanishing problem [48]. Since the basic design is a collection of LSTM cells, Bi-LSTM has the ability to capture bidirectional dependencies within time series data, which can lead to better performance in solving time series data problems.
- BRITS, MRNN, Transformer: According to Cao et al. [12], BRITS uses LSTM as the recurrent component, which utilizes the feature correlations in the unidirectional recurrent dynamical system, and extends this feature-correlated algorithm for the bidirectional case. As illustrated in the study by Yoon et al. [13], MRNN was proposed based on bi-directional RNN to fill the missing data in time series data by referring to the previous and subsequent observations of the time series. Besides, it can operate not only forward and backward within each data stream in the intra-stream directions, but also can operate in the inter-stream directions. Missing data reconstruction using Transformer, on the other hand, is similar to using Transformer for the prediction of energy load. Unlike conventional recurrent neural networks, the self-attention mechanism in Transformer enables the model to weigh the importance of different tokens in sequential data dynamically [49].
- Original context encoder in Iizuka et al. [19]: The original model that was proposed by Ref. Iizuka et al. [19] was also adapted as a baseline model in our experiment. The original network was simply transformed from 2D to 1D, in order to adapt its ability for processing the time series data.

3.4.2. Experimental setting TCN-BiLSTM-CE

After constructing the dataset for training and completing data preprocessing, the next step is to build the TCN-BiLSTM-CE model that we use for missing data reconstruction. Table 3 shows the structure of the generator network in TCN-BiLSTM-CE. Both the global and local discriminators have the same structure. Tables 4 and 5 show the structure of global discriminator and local discriminator respectively, as well as the structure after they are concatenated together. As the data patterns of electrical and heat load data are different, the hyper-parameter tuning was separately done for these two different types of energy data using grid search method. We mainly tuned the two hyper-parameters, learning rate and batch size, using the grid search method, where the learning rate was selected from 0.001, 0.0005, 0.00025 and 0.0001, and the batch size was selected from 4, 8, 16 and 32, respectively. The learning rate and batch size of the network trained on the electricity load dataset are 0.0005 and 8, while the learning rate and batch size of the network trained on the heat load dataset are 0.001 and 16. The model optimizer is chosen as Adam. As mentioned in Section 2.3, T_G and T_D are set to 100 and 50 iterations respectively. λ for calculating network loss in Eq. (7) is set to 0.0004.

As introduced in Section 1, both the electricity load dataset and heat load dataset that were presented in Section 3.1 will be used to validate the effectiveness of our proposed method. The training process randomly creates missing data for 96–264 data points in the original

Table 3
Structure of generator in TCN-BiLSTM-CE.

Type	Kernel size	Dilation rate	Stride	Filter
conv1d.	3	1	1	32
conv1d.	3	1	2	64
conv1d.	3	1	2	128
conv1d.	3	1	2	128
conv1d.	3	1	1	128
conv1d.	3	1	1	128
Bi-LSTM.	3	–	1	128
TCN.	3	1,2,4,8,16	1	128
Bi-LSTM.	3	–	1	128
TCN.	3	1,2,4,8,16	1	128
conv1d.	3	1	1	128
conv1d.	3	1	1	128
deconv1d.	3	1	1/2	128
deconv1d.	3	1	1/2	128
deconv1d.	3	1	1/2	128
conv1d.	3	1	1	64
output	3	1	1	1

Table 4
Structure of global and local discriminator in TCN-BiLSTM-CE.

Type	Kernel size	Stride	Filter	Node
conv1d.	3	2	16	–
conv1d.	3	2	32	–
conv1d.	3	2	64	–
conv1d.	3	2	128	–
FC.	–	–	–	1024

Table 5
Structure of discriminator after concatenating.

Type	Node
concat.	2048
FC.	1

data, corresponding to a total of 4–10 days of missing data in the original data, respectively. By manually creating continuous missing data and discrete missing data in original data, the manually created missing data will be filled by different methods.

4. Results

4.1. Comparison between TCN-BiLSTM-CE and baseline models

In this section, the performance of the proposed TCN-BiLSTM-CE network will be compared with three types of baseline methods, i.e., LSTM methods, SOTA methods, and original image model, as mentioned in Section 3.4.1. The comparison will be done in two ways, i.e., the NRMSE comparison between the reconstructed data and the original, and the visualized comparison between the results of the different methods in a specific time period. The buildings shown in Figs. 2 and 3 in Section 3.1 are used as examples for the visualized comparison for reconstructing continuous missing data up to 10 days and discrete missing data up to 80 %. The selected time periods for electricity and heat load data in result presentation are from '07/08/2017 00:00:00' to '07/11/2017 23:00:00' and '02/20/2024 00:00:00' to '02/23/2024 23:00:00', respectively.

4.1.1. Comparison of different methods with evaluation metrics

The comparison of NRMSE between our proposed TCN-BiLSTM-CE and three other types of methods when reconstructing continuous and discrete missing data in electricity load data and heat load data is shown in Figs. 5–7. From the demonstration, it can be seen that TCN-BiLSTM-CE

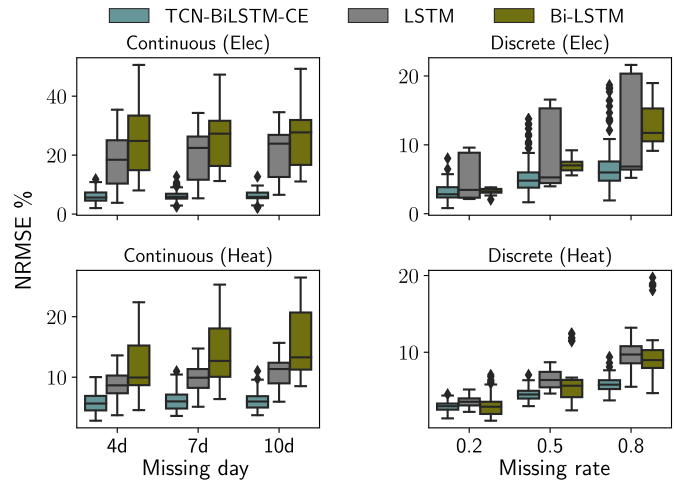

Fig. 5. Comparison of NRMSE between TCN-BiLSTM-CE and LSTM methods for imputing discrete missing in heat load data.

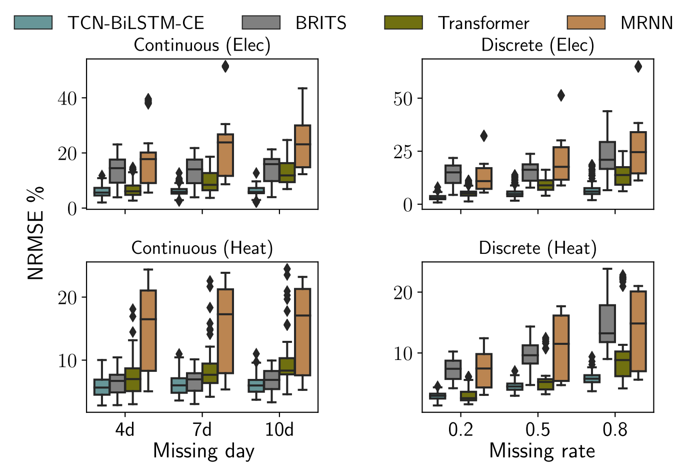

Fig. 6. Comparison of NRMSE between TCN-BiLSTM-CE and SOTAs for imputing continuous and discrete missing in both electricity and heat load data.

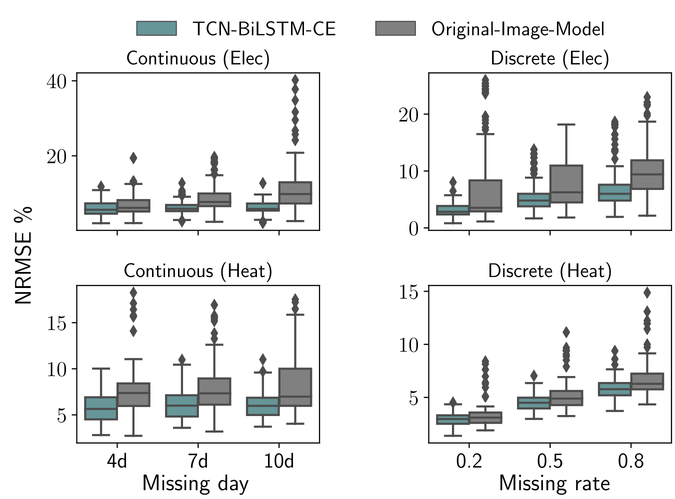

Fig. 7. Comparison of NRMSE between TCN-BiLSTM-CE and original image model for imputing continuous and discrete missing in both electricity and heat load data.

Table 6
Percentage reduction of NRMSE % between TCN-BiLSTM-CE and baseline methods for both continuous and discrete missing data.

Energy	Missing day/rate	LSTM	Bi-LSTM	BRITS	Transformer	MRNN	Image model
Electricity	4	69.54	77.38	61.21	7.58	68.40	8.48
	7	73.76	78.41	58.08	30.10	75.28	23.39
	10	75.57	78.54	63.39	50.72	74.77	40.47
	0.2	18.48	13.11	81.20	44.73	74.09	20.10
	0.5	8.85	31.47	70.42	46.00	72.74	23.18
	0.8	12.55	48.96	71.42	56.49	75.62	36.35
Heat	4	34.72	43.22	15.70	19.25	65.74	23.35
	7	39.51	52.73	13.66	21.82	65.27	18.19
	10	47.31	54.98	12.90	28.29	64.98	14.31
	0.2	16.28	−2.40	59.94	−15.27	56.35	4.26
	0.5	29.53	20.22	53.30	14.21	34.63	8.05
	0.8	40.48	35.64	56.35	34.63	61.10	8.06

significantly outperforms other methods in reconstructing both discrete missing data and continuous missing data for both electricity load data and heat load data. To further compare the effect when reconstructing different patterns of missing data, Table 6 further compares the NRMSE achieved by TCN-BiLSTM-CE with other methods, where the NRMSE used is the median of the NRMSE obtained for all buildings after reconstructing the missing data using different methods in both datasets. Another conclusion that can be drawn by comparing the NRMSE between the different methods for each building is that the gap between the NRMSE obtained using the original image model and TCN-BiLSTM-CE, compared to the LSTM method and the SOTA method, is slightly lower than that of the other two methods, which further illustrates the potential of the cross-domain method for missing data reconstruction in energy load data. Comparisons of the MAE and MBE are shown in Appendix A.

4.1.2. Visualized comparison

In order to further analyze the differences between using different methods, in this section, we visualize and compare the reconstruction results obtained by the different types of methods. We compare the differences between the results obtained by different methods by reconstructing continuous missing data in electrical load data and discrete missing data in heat load data, respectively.

Firstly, Figs. 8 and 9 show the comparison between our proposed method and LSTM methods. From the demonstration we can see, that when reconstructing consecutive missing data in the electricity load data, Bi-LSTM and LSTM do not reconstruct the pattern of decreasing electricity load for the target building, i.e., the significant decrease and increase in electricity load during weekend hours and weekdays respectively. In addition, the estimates tend to converge towards the same value as the length of consecutive missing periods to be reconstructed increases. This is because both Bi-LSTM and LSTM have to use the already estimated value to predict the next missing value, which means that the data obtained by the model are always estimated data with errors. As for the reconstruction of discrete missing data, the demonstration in Fig. 9 also shows a significant difference between two different types of methods. Although the offset of the reconstructed result obtained by LSTM methods is not as large as when reconstructing continuous missing values, the reconstructed data obtained by LSTM and Bi-LSTM appear similar to the phenomenon of adding noise, which is obviously not a desirable result for the purpose of obtaining high-quality data through missing value reconstruction.

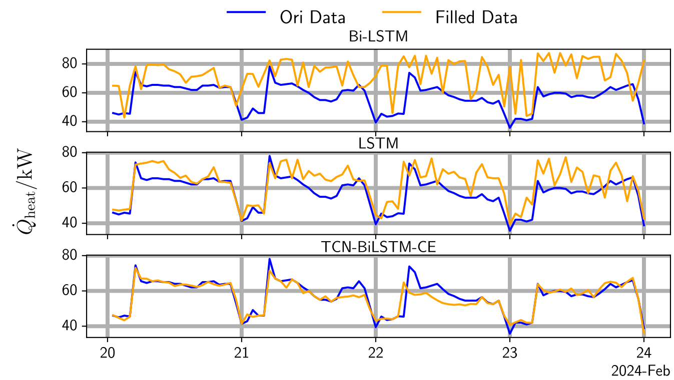

Fig. 9. Comparison of the results of reconstructing the missing data in the heat load data with a discrete missing rate of up to 80%.

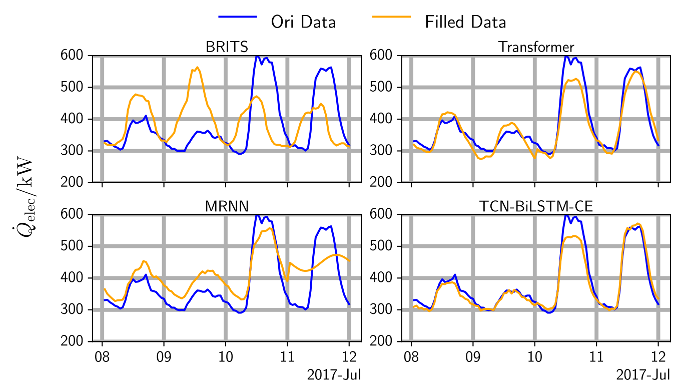

Fig. 10. Comparison of the results of reconstructing in the electricity load data with a continuous missing duration of up to 10 days.

Figs. 10 and 11 show the comparison between TCN-BiLSTM-CE and SOTAs. Both MRNN and BRITS appear to produce a noise-like results when reconstructing missing data, which also explains the higher NRMSE in Table 6. Transformer, on the other hand, although the reconstruction results are much better than the other two SOTAs, shows a greater offset as well as less smooth results compared to the original data when compared to TCN-BiLSTM-CE.

Finally, the comparison between original image model and TCN-BiLSTM-CE is shown in Figs. 12 and 13. From the presentation of the results, we can see the smoothness when reconstructing discrete missing data, TCN-BiLSTM-CE is better than the direct application of the image model, which is also in line with the conclusions presented in Fig. 7 and Table 6.

Fig. 14 compares the original data with data filled by our proposed method for the electricity and heat data. The comparison is made for the continuous missing scenario (10 continuous missing days) and the

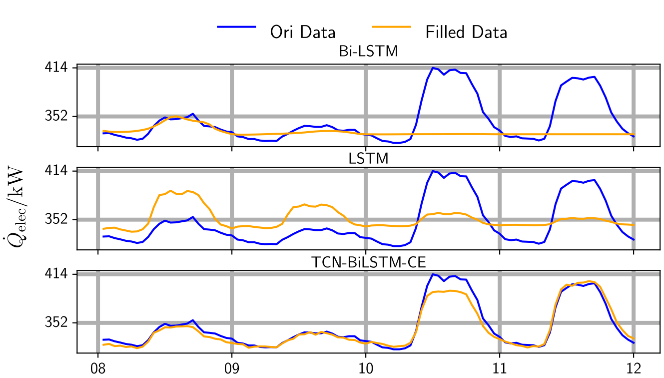

Fig. 8. Comparison of the results of reconstructing in the electricity load data with a continuous missing duration of up to 10 days.

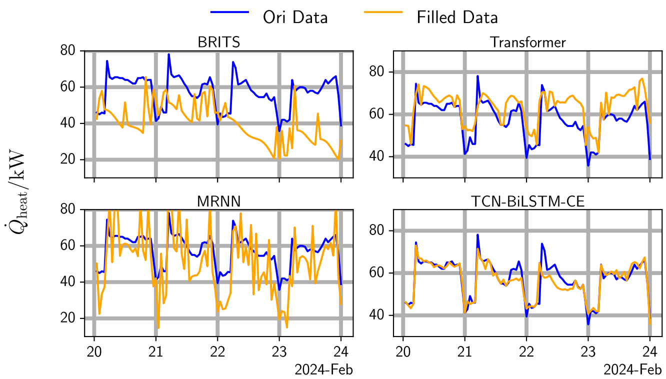

Fig. 11. Comparison of the results of reconstructing in the heat load data with a discrete missing rate of up to 80%.

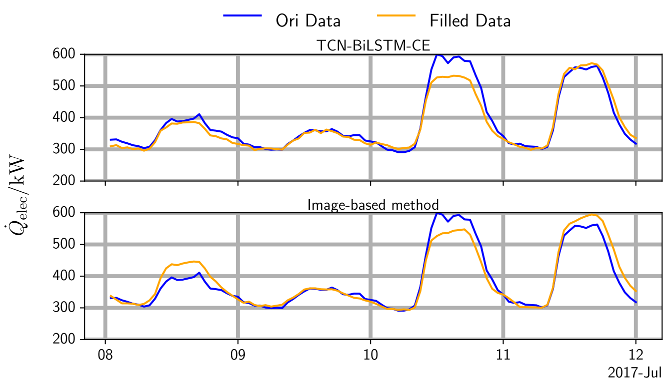

Fig. 12. Comparison of the results of reconstructing in the electricity load data with a continuous missing duration of up to 10 days.

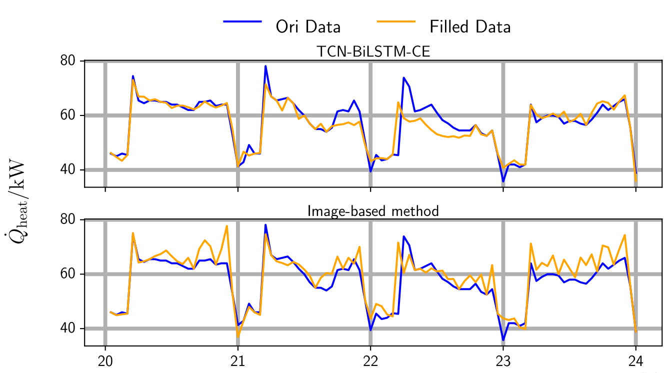

Fig. 13. Comparison of the results of reconstructing in the heat load data with a discrete missing rate of up to 80%.

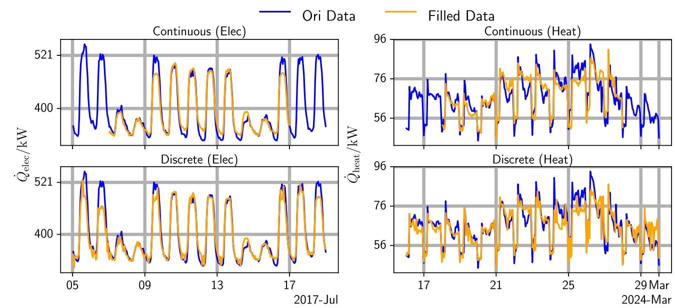

Fig. 14. Comparison of the results of reconstructing the continous and discrete missing data for both electricity heat data with our method.

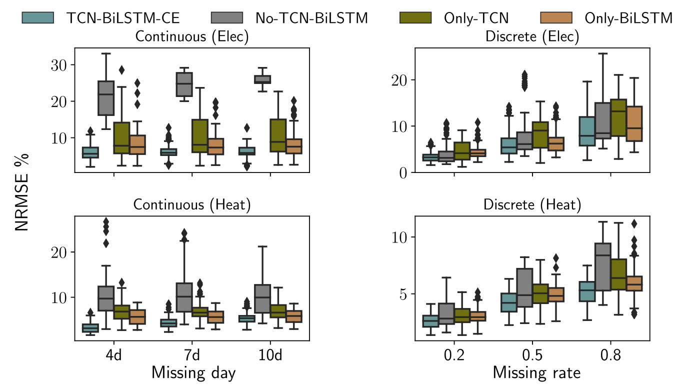

Fig. 15. Comparison of the NRMSE for the ablation study.

discrete missing scenario (80 % missing rate). The results show that our method is effective at filling both continuous and discrete missing values for different energy types. It also produces good reconstructions of the trend changes presented by the heat load data.

4.2. Ablation study

Having compared the model performance between TCN-BiLSTM-CE and other baseline methods, the next part is to figure out how different modules affect model performance. Therefore, we conducted an ablation study, removing different modules from the network and comparing the performance of different network configurations. To prove the role of TCN and Bi-LSTM in optimising model performance, we set up four different network configurations for ablation study: TCN only, Bi-LSTM only, both TCN and Bi-LSTM, and neither TCN nor Bi-LSTM. Fig. 15 shows the comparison between the original and filled data with the metrics that have been used in Section 4.1. Across all ablation scenarios, the complete TCN-BiLSTM-CE model delivers the most accurate reconstructions, achieving the lowest MAE and NRMSE and retaining near-zero MBE when filling continuous gaps in both electricity and heat load data. For discrete missing intervals, the Bi-LSTM-only variant nearly matches the full model's error rates across all missing-rate settings, yet its bias drifts noticeably under- or overestimating the true values, whereas the full model stays unbiased under every condition. Comparatively, the Bi-LSTM-only architecture outperforms the TCN-only variant on every metric, underscoring that leveraging temporal information from both before and after the missing time range is critical for filling the missing data. Table 7 shows the relative reduction in NRMSE for the different ablation scenarios (only-TCN, only-BiLSTM and no-TCN-BiLSTM) compared to our proposed method, to better compare their effectiveness. The results of the comparison for MAE and MBE, are shown in Appendix B.

4.3. Comparison between different training procedures

As described in Section 2.3, in order to verify whether our proposed training procedure, i.e., using the pre-trained generator and discriminator to perform the adversarial training, is the optimized training procedure, in this section, we compare its results with two other training procedures: training the generator and discriminator simultaneously from scratch, and training the generator only. Fig. 16 shows the comparison of NRMSE. As can be seen from the comparisons shown in Fig. 16, training the generator and discriminator simultaneously is significantly worse than the other two training methods. More outliers

Table 7
Percentage reduction of NRMSE % between different ablation configurations.

Energy	Missing day/rate	No-TCN-BiLSTM	Only-TCN	Only-BiLSTM
Electricity	4d	73.16	40.68	26.83
	7d	74.84	38.53	20.58
	10d	76.14	40.62	23.98
	0.2	14.59	26.85	23.63
	0.5	18.92	25.74	5.74
	0.8	19.75	24.03	15.09
Heat	4d	67.93	52.28	42.95
	7d	58.41	35.76	22.25
	10d	49.59	21.85	6.08
	0.2	22.41	15.62	15.12
	0.5	16.87	16.87	14.16
	0.8	32.40	21.96	12.15

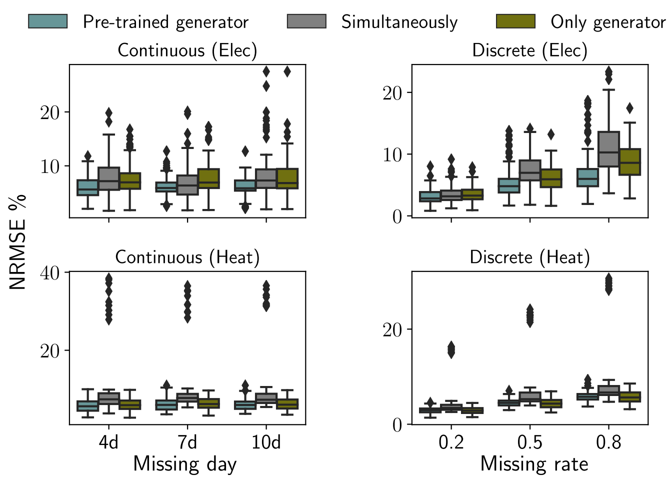

Fig. 16. Comparison of NRMSE between different training procedure.

in the demonstration also suggest that training the generator and discriminator simultaneously is less stable than the other two methods. Training only the generator, although the results are not far from our proposed training procedure in missing data reconstruction for heat load data, shows significantly larger gap when it is used for electric load data. On the other hand, although training only the generator shows few differences compared to our proposed training procedure on heat load data, the gap increases significantly when used for electric load data. To further compare the differences when reconstructing missing data for different patterns, Table 8 lists the NRMSE differences between other two training procedures and our proposed training procedure in Section 2.3. Two other criteria for evaluating the performance of different training methods are training time and peak memory usage, which are presented in Table 9. The other two simpler training processes require less training time and consume less peak memory during training. However, even though the training process we proposed in Section 2.3 takes the longest time and requires the highest peak memory, the peak memory occupied by our proposed method is not that high relative to other deep learning models that also run based on the CPU [50,51], while the performance improvement over the other two methods is significant. The results of the comparisons of the other two indicators, MAE and MBE, are shown in Appendix C.

Table 8
Percentage reduction of NRMSE between different training procedures.

Energy	Missing day/rate	simultaneously	Only generator
Electricity	4d	21.57	19.06
	7d	7.32	14.61
	10d	19.84	14.21
	0.2	10.76	13.98
	0.5	31.08	18.89
	0.8	41.72	30.28
Heat	4d	23.99	4.76
	7d	22.72	3.56
	10d	18.41	1.36
	0.2	13.51	3.40
	0.5	14.75	3.59
	0.8	13.56	2.45

Table 9
Time and peak memory usage among different training procedures.

Procedure	Time usage in s	Peak memory usage in MB
Pre-trained generator	438	18.86
Simultaneously	182	7.45
Only generator	225	3.64

4.4. Comparison between univariate and multivariate data as input

Another part worth exploring is the effect of different input data on the TCN-BiLSTM-CE model. Although the observed data before and after the missing data provide contextual information for reconstructing the missing data through contextual inference, if the continuous missing time length is too long or the discrete missing rate is too large, inference solely through the observed data before and after missing time periods is likely to result in a reconstruction that ignores the pattern of the original energy load data. As described in Section 2.1, the input data to our model includes, in addition to the energy load data, time variables and weather variables that may affect the pattern of the energy load data, thus providing additional context to reconstruct the pattern of the missing time periods. In this section, we compare this approach with the results of using only energy load data as input.

The comparison of NRMSE between reconstructed and original data when the input data are univariate and multivariate, respectively, is shown in Fig. 17. Demonstration of Fig. 17 shows the superior performance of using multivariate data as input compared to using univariate data as input, which is further illustrated by comparing the reduction in NRMSE using different inputs in Table 10.

Visualized comparisons also demonstrate better performance when using multivariate data as input to reconstruct missing data. Figs. 18 and 19 present the comparisons when reconstruction is performed for continuous missing electric load data and discrete missing heat load

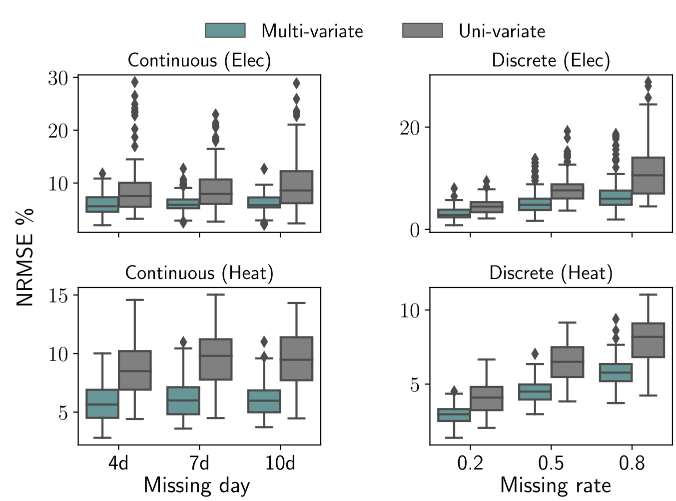

Fig. 17. Comparison of NRMSE between different training procedure.

Table 10
Percentage reduction of NRMSE % between different variables as input.

Energy	Missing day/rate	Uni-variate
Electricity	4d	25.74
	7d	25.92
	10d	32.12
	0.2	36.64
	0.5	37.10
	0.8	43.38
Heat	4d	33.52
	7d	38.79
	10d	36.86
	0.2	27.57
	0.5	30.87
	0.8	29.42

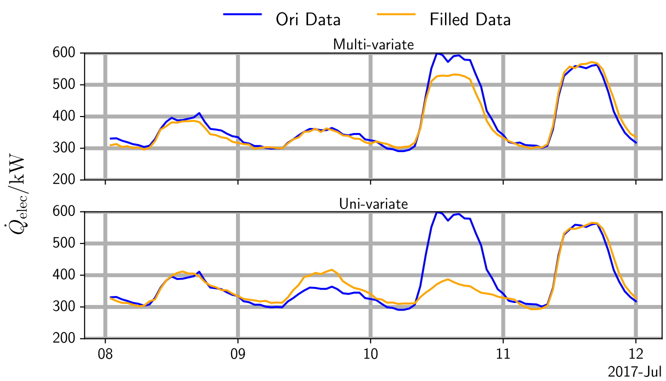

Fig. 18. Comparison of results for reconstructing continuous missing data in electrical load data using multi-variate and uni-variate data as input.

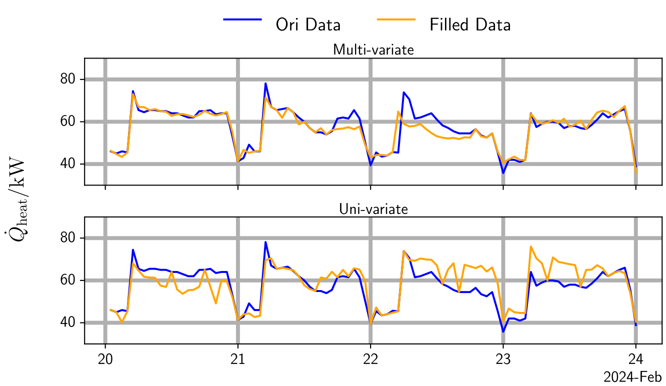

Fig. 19. Comparison of results for reconstructing discrete missing data in heat load data using multi-variate and uni-variate data as input.

data, respectively. It can be observed that reconstructing continuous missing data with only univariate data as input does not reconstruct the pattern of the original data completely and accurately, and the bias in load pattern reconstruction is also reflected in the reconstruction of discrete missing data, resulting in more noise when reconstructing discrete missing data. Comparisons of MAE and MBE are shown in Appendix D.

5. Conclusion

This paper proposed a context encoder framework called TCN-BiLSTM-CE for missing energy data reconstruction based on the contextual information of the observed data before and after the missing part and other conditional variables such as temperature, day of week, and hour, which shows a significant optimization of the performance. The TCN-BiLSTM-CE framework demonstrates the potential of cross-domain approaches for energy data analysis. By integrating the TCN and Bi-LSTM modules, the method effectively adapts to time-series data, helping to improve feature extraction and thus increasing its effectiveness in reconstructing missing energy data. The results revealed the beneficial role of our proposed framework, leading to better performance than other baseline methods commonly used in missing energy data imputation. The results demonstrate that relative to traditional LSTM methods (LSTM, Bi-LSTM), SOTAs (BRITS, MRNN, Transformer) and original image model, NRMSE reductions are achieved using TCN-BiLSTM-CE for 8.85%–78.54%, 7.58%–81.20% and 4.26%–40.47% of the NRMSE decreases. Moreover, compared to the percentage reduction in NRMSE achieved using SOTA (BRITS, MRNN, Transformer) and LSTM methods (LSTM, Bi-LSTM), the percentage reduction in NRMSE of the results obtained using the original image model is smaller, which also proves that the performance of the original image model is slightly better than those other two approaches, thus providing a good evidence of the potential for using a cross-domain approach for missing data reconstruction in energy data.

In future work, the potential of using TCN-BiLSTM-CE network for missing data reconstruction in real-time applications will be further explored. Using pre-trained models for one building to reconstruct missing data for energy load data for buildings with the same energy use pattern, through transfer learning, is likewise a worthwhile focus of work.

CRediT authorship contribution statement

Mengbo Yu: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Alexander Neubauer:** Writing – review & editing. **Stefan Brandt:** Project administration. **Martin Kriegel:** Supervision, Resources.

Declaration of generative AI in scientific writing

During the preparation of this work the author(s) used DeepL to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Comparison of MAE and MBE between TCN-BiLSTM-CE and other baseline methods

Figs. A.20–A.25 compare MAE and MBE for our proposed TCN-BiLSTM-CE against the three baseline methods introduced in Section 3.4.1 when reconstructing both continuous and discrete missing data in electricity and heat load data. Consistent with the NRMSE results discussed in Section 4.1.1, our method achieves markedly lower MAE and maintains MBE values close to zero, demonstrating both reduced reconstruction error and an ability to avoid systematic over- or underestimation of the missing data.

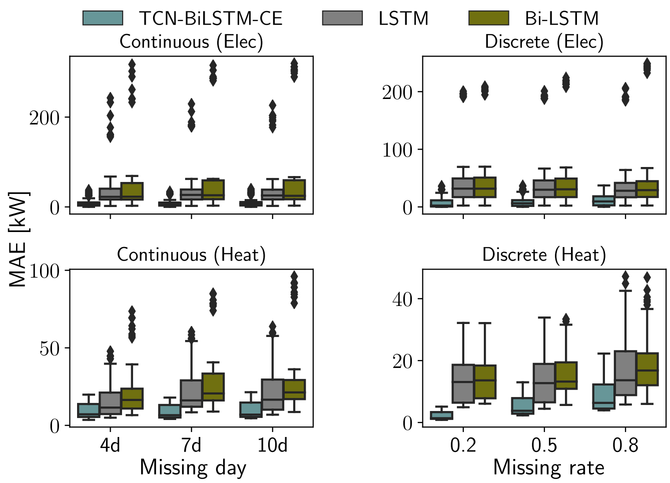

Fig. A.20. Comparison of MAE between TCN-BiLSTM-CE and LSTM methods for imputing discrete missing in heat load data.

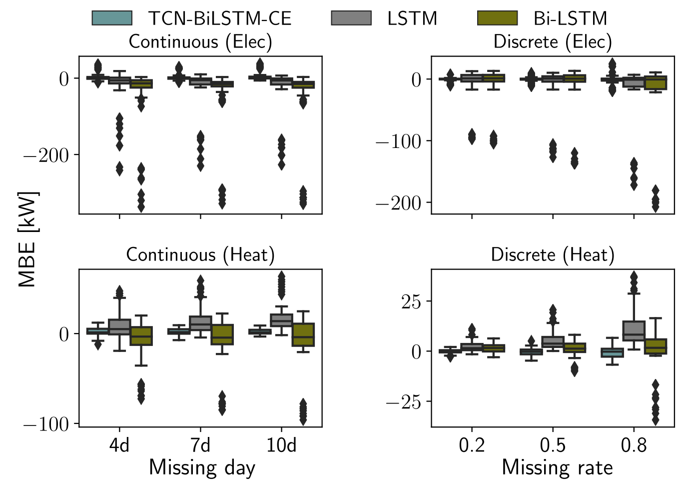

Fig. A.23. Comparison of MBE between TCN-BiLSTM-CE and LSTM methods for imputing discrete missing in heat load data.

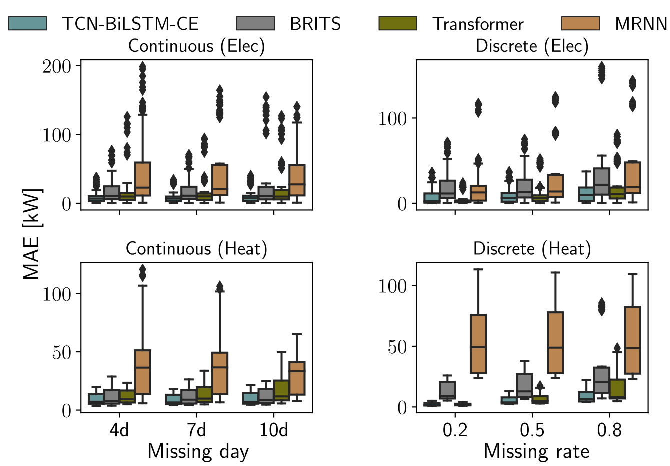

Fig. A.21. Comparison of MAE between TCN-BiLSTM-CE and SOTAs for imputing continuous and discrete missing in both electricity and heat load data.

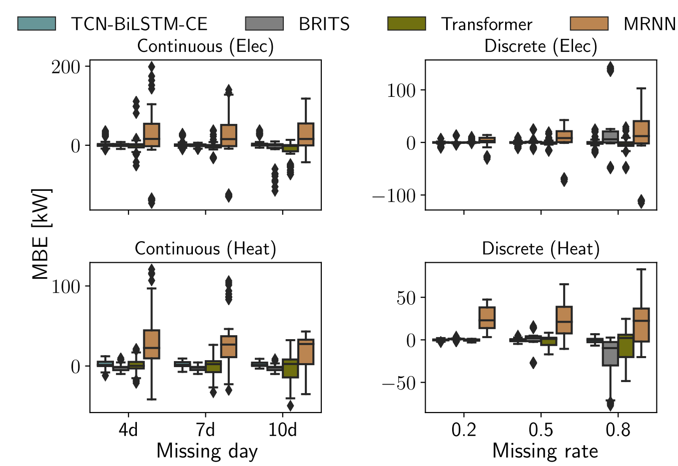

Fig. A.24. Comparison of MBE between TCN-BiLSTM-CE and SOTAs for imputing continuous and discrete missing in both electricity and heat load data.

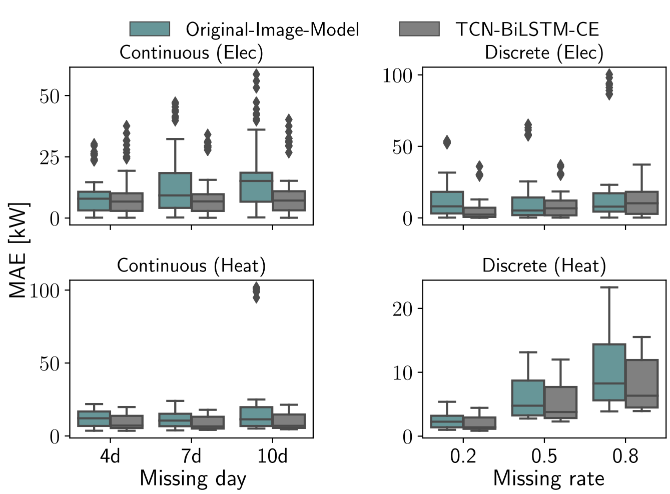

Fig. A.22. Comparison of MAE between TCN-BiLSTM-CE and original image model for imputing continuous and discrete missing in both electricity and heat load data.

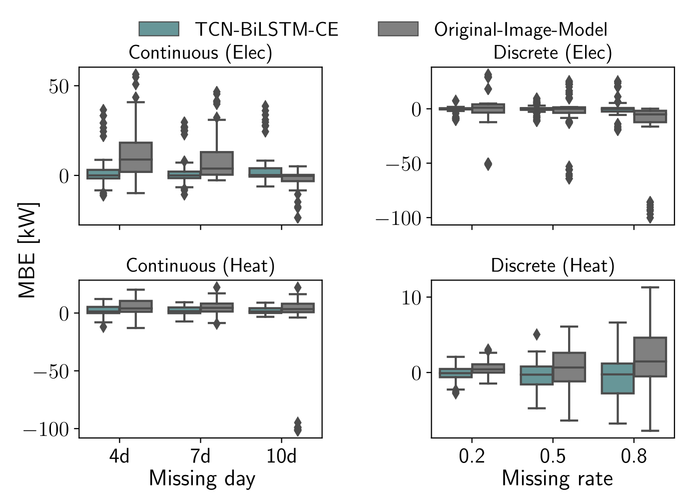

Fig. A.25. Comparison of MBE between TCN-BiLSTM-CE and original image model for imputing continuous and discrete missing in both electricity and heat load data.

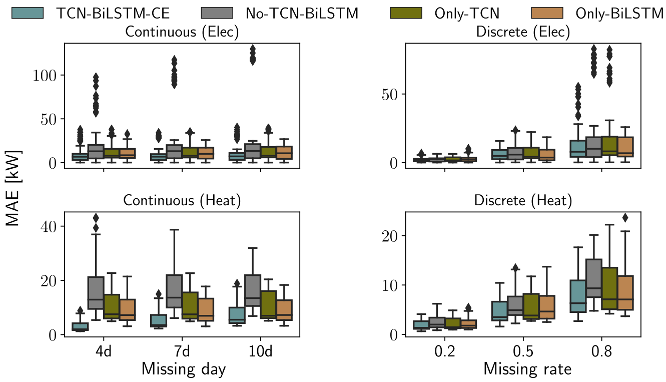

Fig. B.26. Comparison of MAE between different ablation settings.

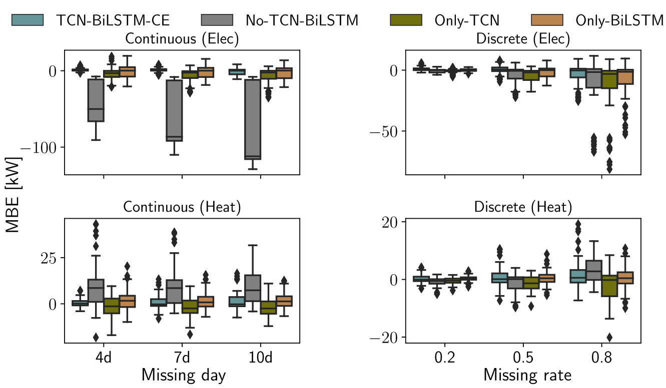

Fig. B.27. Comparison of MBE between different ablation settings.

Appendix B. Comparison of MAE and MBE for the ablation study

Figs. B.26 and B.27 present MAE and MBE across different ablation scenarios. These results corroborate the NRMSE findings reported in Section 4.2: TCN-BiLSTM-CE yields the lowest MAE and maintains MBE closest to 0 compared to other ablation scenarios. Moreover, as in Section 4.2, the BiLSTM-only setting surpasses the TCN-only setting on both metrics, further highlighting the role of the Bi-LSTM in this network.

Appendix C. Comparison of MAE and MBE between different training procedures

The comparison of MAE and MBE between different training procedures is shown in Figs. C.28 and C.29. The results show the same conclusion as from the comparison of NRMSE between different training procedures in Section 4.3 with a lower MAE and a near-zero MBE. Also similar to the results from the Section 4.3, training only the generator is better than simultaneously training both generator and discriminator, which further illustrates the importance of initially training the generator, as described in Section 2.3.

Appendix D. Comparison of MAE and MBE between univariate and multivariate data as input

Figs. D.30 and D.31 compare the performance of our model between using multivariate data and univariate data as input. Consistent with the conclusion in Section 4.4, using multivariate data as input achieves a lower MAE and an MBE closer to zero, further demonstrating that incorporating multiple input features leads to more accurate and unbiased reconstructions than relying on only one variable.

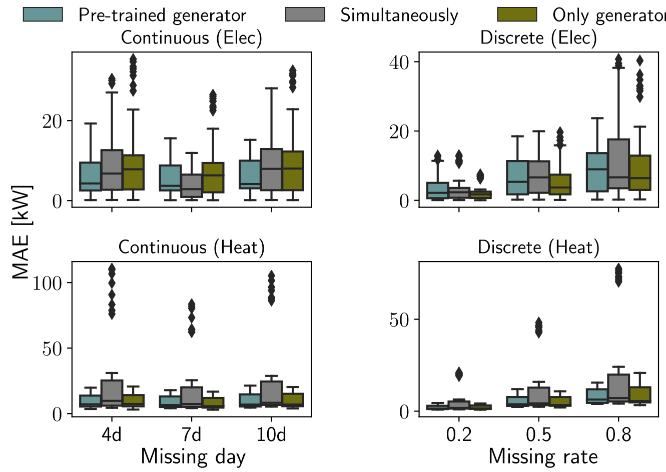

Fig. C.28. Comparison of MAE between different training procedures.

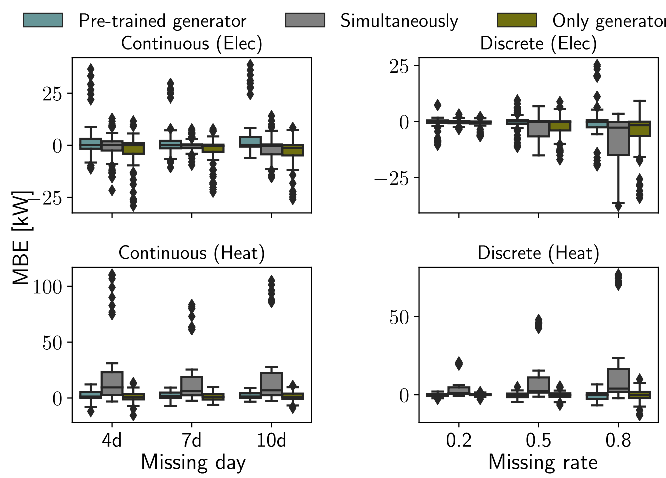

Fig. C.29. Comparison of MBE between different training procedures.

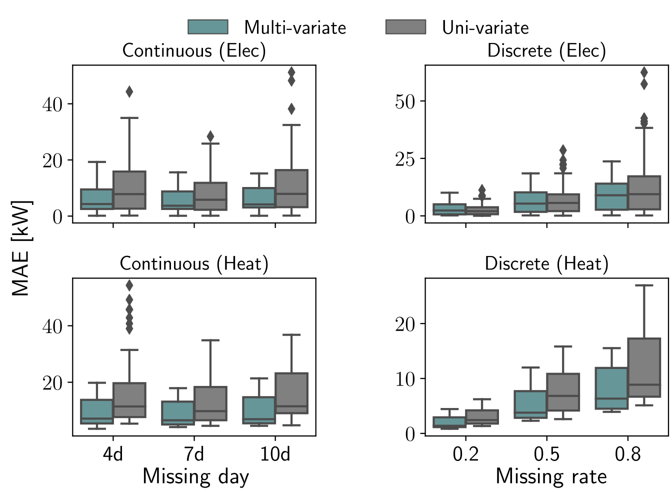

Fig. D.30. Comparison of MAE between using uni-variate and multi-variate data as input.

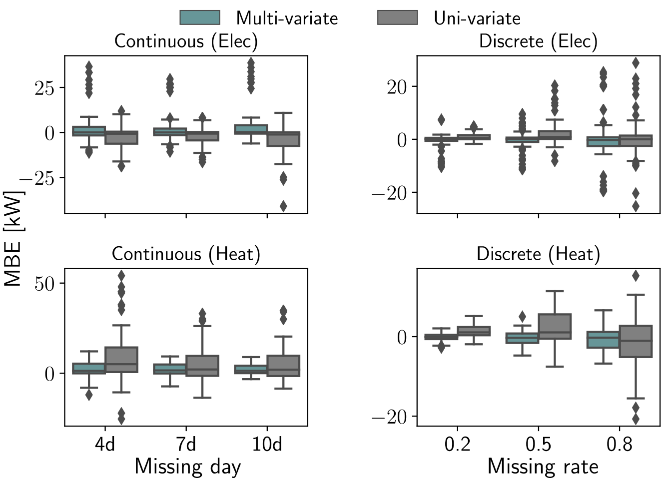

Fig. D.31. Comparison of MBE between using uni-variate and multi-variate data as input.

Data availability

The code and electrical load data used in this paper are freely available at https://github.com/yu9731/TCN-BiLSTM-CE.

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